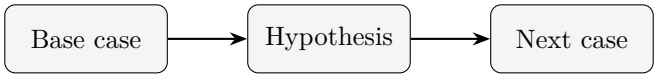
DISCRETE STRUCTURES

CMSC250

Chapter 14: Mathematical Induction

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Base cases, inductive hypotheses, and rigorous proof steps



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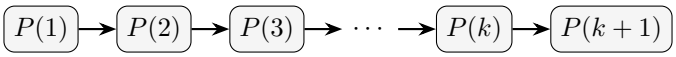
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1 Why induction is needed

Many claims concern every integer n from a starting point onward. Checking $n = 1, 2, 3$ gives evidence, but never proves an infinite collection of statements. **Mathematical induction** proves a starting statement and then proves that each true statement forces the next one to be true.

Key idea. Induction has two separate obligations: prove $P(n_0)$, then prove $P(k) \Rightarrow P(k+1)$ for an arbitrary integer $k \geq n_0$. Neither obligation may be skipped.



Think of dominoes: the base case tips the first domino, and the induction step establishes the mechanism that tips the next domino. The metaphor is not a substitute for writing the proof.

2 A complete induction template

To prove $P(n)$ for every integer $n \geq n_0$:

1. **Define $P(n)$.** Write the exact statement being proved, including its domain.
2. **Base case.** Substitute $n = n_0$ and verify the equality or assertion on both sides.
3. **Inductive hypothesis (IH).** Let $k \geq n_0$ be arbitrary and *assume* $P(k)$.
4. **Inductive step.** State explicitly what $P(k+1)$ says; derive it using the IH, with any other needed facts justified.
5. **Conclusion.** Invoke the Principle of Mathematical Induction and finish with ■.

3 Sums: from k terms to $k+1$ terms

Worked example: sum of the first n positive integers

Question. Prove that for every integer $n \geq 1$, $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.

Proof. Let $P(n)$ be the displayed identity. **Base case:** When $n = 1$, the left side is 1 and the right side is $1(2)/2 = 1$, so $P(1)$ holds. **Inductive hypothesis:** Let $k \geq 1$ be an arbitrary integer. Assume $P(k)$, that is, $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$. **Inductive step:** We must prove $\sum_{i=1}^{k+1} i = \frac{(k+1)(k+2)}{2}$. Starting with the left side,

$$\begin{aligned}
\sum_{i=1}^{k+1} i &= \left(\sum_{i=1}^{k} i\right) + (k+1) && \text{split off last term} \\
&= \frac{k(k+1)}{2} + (k+1) && \text{by IH} \\
&= \frac{(k+1)(k+2)}{2} && \text{factor and simplify.}
\end{aligned}$$

Thus $P(k+1)$ holds. By mathematical induction, the identity holds for all integers $n \geq 1$. ■

Why the proof works. The only use of the IH is to replace the *first k terms*. Writing the target before doing algebra prevents accidentally proving the wrong formula.

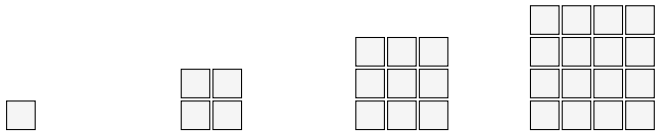
Worked example: sum of the first n odd positive integers

Question. Prove $\sum_{i=1}^{n}(2i-1) = n^2$ for all integers $n \geq 1$.

Proof. Define $P(n)$ by the given equality. **Base:** $n = 1$ gives $1 = 1^2$. **IH:** Fix arbitrary $k \geq 1$ and assume $\sum_{i=1}^{k}(2i-1) = k^2$. **Step:**

$$\sum_{i=1}^{k+1}(2i-1) = k^2 + (2(k+1)-1) = k^2 + 2k + 1 = (k+1)^2.$$

The first equality uses the IH on the first k terms. Therefore $P(k+1)$ holds and the claim follows by induction. ■



1^2 2^2 3^2 4^2

4 Divisibility proofs

Recall: $d \mid m$ means $m = dt$ for some integer t. An induction proof of divisibility must produce an integer multiple of d in the step.

Worked example: a power expression divisible by 3

Question. Prove that $3 \mid (4^n - 1)$ for every integer $n \geq 0$.

Proof. Let $P(n)$ assert that $3 \mid (4^n - 1)$. **Base:** At $n = 0$, $4^0 - 1 = 0 = 3 \cdot 0$, so $P(0)$ holds. **IH:** Fix an arbitrary integer $k \geq 0$ and suppose $4^k - 1 = 3t$ for some integer t. **Step:**

$$\begin{aligned} 4^{k+1} - 1 &= 4 \cdot 4^k - 1 \\ &= 4(4^k - 1) + 3 \\ &= 4(3t) + 3 = 3(4t+1). \end{aligned}$$

Since $t \in \mathbb{Z}$ and the integers are closed under multiplication and addition, $4t + 1 \in \mathbb{Z}$. Therefore $3 \mid (4^{k+1} - 1)$. The claim follows by induction. ■

Worked example: a product of consecutive integers

Question. Prove $6 \mid (n^3 - n)$ for every integer $n \geq 0$ using induction.

Proof. Let $P(n)$ be $6 \mid (n^3 - n)$. **Base:** $n = 0$ gives 0, divisible by 6. **IH:** Fix arbitrary $k \geq 0$ and assume $k^3 - k = 6t$ for some integer t. **Step:**

$$\begin{aligned}(k+1)^3 - (k+1) &= (k^3 - k) + 3k(k+1) \\ &= 6t + 3k(k+1).\end{aligned}$$

One of $k, k+1$ is even, so $k(k+1) = 2s$ for some integer s. Hence the expression equals $6(t+s)$, an integer multiple of 6. Induction completes the proof. ■

5 Inequalities: watch the direction

Worked example: an exponential eventually dominates a linear term

Question. Prove $2^n \geq n + 1$ for every integer $n \geq 0$.

Proof. Define $P(n)$ by $2^n \geq n + 1$. **Base:** $2^0 = 1 \geq 1$. **IH:** Let $k \geq 0$ be arbitrary and assume $2^k \geq k + 1$. **Step:** Multiplying by the positive number 2 preserves the inequality:

$$2^{k+1} = 2 \cdot 2^k \geq 2(k+1) \geq (k+1) + 1,$$

where the last inequality holds because $k \geq 0$. Thus $P(k+1)$ holds, so the claim follows by induction. ■

Worked example: a factorial lower bound

Question. Prove $n! \geq 2^{n-1}$ for every integer $n \geq 1$.

Proof. Base: $1! = 1 = 2^0$. **IH:** Fix arbitrary $k \geq 1$ and assume $k! \geq 2^{k-1}$. **Step:**

$$(k+1)! = (k+1)k! \geq (k+1)2^{k-1} \geq 2 \cdot 2^{k-1} = 2^k,$$

since $k + 1 \geq 2$. The statement follows by induction. ■

6 Starting at a different index

If a claim begins at $n = 4$, the base case is $n = 4$, not $n = 0$ or $n = 1$. The IH must likewise say $k \geq 4$.

Worked example: a bound valid from $n = 4$ onward

Question. Prove $2^n \geq n^2$ for every integer $n \geq 4$.

Proof. Base: $2^4 = 16 = 4^2$. **IH:** Let $k \geq 4$ be arbitrary and suppose $2^k \geq k^2$. **Step:**

$$2^{k+1} \geq 2k^2 \geq (k+1)^2,$$

because $2k^2 - (k+1)^2 = k^2 - 2k - 1 \geq 0$ for $k \geq 4$ (indeed, $k^2 - 2k - 1 = (k-1)^2 - 2 \geq 7$). Thus the induction step holds. ■

7 Common proof-writing mistakes

- **Assuming what must be proved:** $P(k+1)$ is the target, never the IH.
- **Skipping the base:** the implication $P(k) \Rightarrow P(k+1)$ alone does not start the chain.
- **Using the IH on the wrong expression:** rewrite a $k+1$ sum as the first k terms plus the last term.
- **Leaving divisibility as an unexplained congruence:** show a concrete integer multiple when proving from the definition.
- **Not checking inequality bounds:** the extra comparison may require $k \geq n_0$.

8 Practice questions with complete solutions

Question. 1. Prove $\sum_{i=1}^{n} 3i = \frac{3n(n+1)}{2}$ for $n \geq 1$.

Solution. Base $n = 1$: both sides equal 3. Suppose the formula holds at arbitrary $k \geq 1$. Then

$$\sum_{i=1}^{k+1} 3i = \frac{3k(k+1)}{2} + 3(k+1) = \frac{3(k+1)(k+2)}{2}.$$

By induction, the formula holds for every $n \geq 1$. ■**Question.** 2. Prove $5 \mid (6^n - 1)$ for all integers $n \geq 0$.

Solution. Base $n = 0$: $6^0 - 1 = 0$. Suppose $6^k - 1 = 5t$ for some integer t and arbitrary $k \geq 0$. Then

$$6^{k+1} - 1 = 6(6^k - 1) + 5 = 5(6t + 1).$$

Since $6t+1$ is an integer, the desired divisibility follows, and induction completes the proof. ■**Question.** 3. Prove $\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$ for $n \geq 1$.

Solution. At $n = 1$, $2 = 1 \cdot 2 \cdot 3/3$. Assume the identity at arbitrary $k \geq 1$. Adding the $(k+1)$st term gives

$$\begin{aligned}\sum_{i=1}^{k+1} i(i+1) &= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2) \\ &= \frac{(k+1)(k+2)(k+3)}{3}.\end{aligned}$$

The target is obtained, so the formula follows by induction. ■